

Amazon SNS to SQS Integration

To Begin with the Lab

Summary of the Lab

In this lab, **Amazon SNS** and **Amazon SQS** were integrated so that an SNS topic could send messages directly to an SQS queue. The lab first created a standard SNS topic and a standard SQS queue using default settings. Then the SQS queue was added as a subscription to the SNS topic. The main learning point was the **SQS access policy**, which must explicitly allow the SNS topic to send messages to the queue. After updating the queue policy with the correct **root user ARN**, **SQS queue ARN**, and **SNS topic ARN**, the integration became valid. The video explained that this setup is useful because SQS does not accept messages from SNS unless the access policy allows it.

Understanding SNS to SQS Integration

This integration exists so that SNS can fan out messages to SQS, allowing one topic to deliver notifications to a queue for later processing. It is useful when you want decoupled communication between a publisher and a consumer. In this case, SNS acts as the message broadcaster and SQS becomes the subscriber. The important part is that SQS will not accept messages from SNS unless the queue policy explicitly allows the SNS topic ARN.

Example:

A system publishes an alert to an SNS topic. That topic forwards the alert to an SQS queue. A backend application can later read the message from SQS and process it asynchronously.

Lab Steps

Step 1 – Create the SNS Topic

- Sign in to the AWS Management Console.
- Open **Amazon SNS**.
- Go to **Topics**.
- Click **Create topic**.
- Select **Standard**.
- Enter the topic name: Test-Topic

Create topic

Details

Type [Info](#)
Topic type cannot be modified after topic is created.

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ **Standard**

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

Name
Test-Topic
Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).

Display name - optional [Info](#)
To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.
My Topic
Maximum 100 characters.

► **Encryption - optional**
Amazon SNS provides in-transit encryption by default. Enabling server-side encryption adds at-rest encryption to your topic.

► **Access policy - optional** [Info](#)
This policy defines who can access your topic. By default, only the topic owner can publish or subscribe to the topic.

- Leave the default options unchanged.
- Click **Create topic**.

Step 2 – Create the SQS Queue

- Open **Amazon SQS**.
- Go to **Queues**.
- Click **Create queue**.
- Select **Standard** queue type.
- Enter the queue name: Test-Queue

Create queue

Details

Type
Choose the queue type for your application or cloud infrastructure.

☒ **Standard** [Info](#)

- At-least once delivery, message ordering isn't preserved
- At-least once delivery
- Best-effort ordering

☐ FIFO [Info](#)

- First-in-first-out delivery, message ordering is preserved
- First-in-first-out delivery
- Exactly-once processing

ⓘ You can't change the queue type after you create a queue.

Name
Test-Queue
A queue name is case-sensitive and can have up to 80 characters. You can use alphanumeric characters, hyphens (-), and underscores (_).

Configuration [Info](#)
Set the maximum message size, visibility to other consumers, and message retention.

Visibility timeout [Info](#)
30 Seconds
Should be between 0 seconds and 12 hours.

Message retention period [Info](#)
4 Days
Should be between 1 minute and 14 days.

Delivery delay [Info](#)
0 Seconds

Maximum message size [Info](#)
1024 KiB

- Keep the default settings.
- Click **Create queue**.

Step 3 – Add the SQS Queue as an SNS Subscription

- Open the SNS topic.
- Click **Create subscription**.
- Select **SQS** as the protocol.
- Choose the SQS queue created earlier.

Amazon SNS > Subscriptions > Create subscription

Create subscription

Details

Topic ARN
arn:aws:sns:us-east-1:463646775279:Test-Topic

Protocol
The type of endpoint to subscribe
Amazon SQS

Endpoint
Only Amazon SQS standard queues will be listed and can receive notifications from an Amazon SNS standard topic.
arn:aws:sqs:us-east-1:463646775279:Test-Queue

☐ Enable raw message delivery

After your subscription is created, you must confirm it.

► **Subscription filter policy - optional** Info
This policy filters the messages that a subscriber receives.

► **Redrive policy (dead-letter queue) - optional** Info
Send undeliverable messages to a dead-letter queue.

- Create the subscription.

Step 4 – Update the SQS Access Policy

- Open the SQS queue.
- Go to the **Access policy** section.
- Click **Edit**.

Amazon SQS > Queues > Test-Queue

Details

Name Test-Queue	Type Standard	ARN arn:aws:sqs:us-east-1:463646775279:Test-Queue
Encryption Amazon SQS key (SSE-SQS)	URL https://sqs.us-east-1.amazonaws.com/463646775279/Test-Queue	Dead-letter queue -

► More

Queue policies | Monitoring | SNS subscriptions | Lambda triggers | EventBridge Pipes | Dead-letter queue | Tagging | Encryption | Dead-letter queue redrive tasks

Access policy Info

Define who can access your queue.

```
{
  "Version": "2012-10-17",
  "Id": "__default_policy_ID",
  "Statement": [
    {
      "Sid": "__owner_statement",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam:463646775279:root"
      },
      "Action": "sqs:*"
    }
  ]
}
```

Edit

- Start with the existing policy that allows only the root user.

- Update the JSON policy to include:
 - the root user ARN
 - the SQS queue ARN
 - the SNS topic ARN

The screenshot shows the AWS IAM console interface for editing a policy named 'Test-Queue'. The 'Access policy' section is highlighted with a red box, displaying the following JSON policy:

```

13  {
14    "Effect": "Allow",
15    "Principal": {
16      "Service": "sns.amazonaws.com"
17    },
18    "Action": "sqs:SendMessage",
19    "Resource": "arn:aws:sqs:us-east-1:123456789012:Test-Queue",
20    "Condition": {
21      "ArnLike": {
22        "aws:SourceArn": "arn:aws:sns:us-east-1:123456789012:Test-Topic"
23      }
24    }
25  }
26 }
27
28  JSON
  
```

Below the policy editor, the 'Redrive allow policy' section is visible, which is optional and used to identify source queues that can use the queue as a dead-letter queue.

- Save the updated policy.

Step 5 – Verify the Integration

- Return to the SNS topic.
- Publish a test message.
- Message subject: CloudFreeks Test

The screenshot shows the AWS SNS console interface for publishing a message to a topic. The 'Publish message to topic' form is displayed, with the 'Subject - optional' field highlighted by a red box and containing the text 'CloudFreeks Test'. The 'Message details' section shows the 'Topic ARN' and 'Message group ID' fields. The 'Message body' section is also visible, showing the 'Message structure' and 'Message body to send to the endpoint' options.

- Message body: Testing Message From SNS

This setting applies only to mobile application endpoints. The number of seconds that the push notification service has to deliver the message to the endpoint.

Message body

Message structure

☒ Identical payload for all delivery protocols. The same payload is sent to endpoints subscribed to the topic, regardless of their delivery protocol.

☐ Custom payload for each delivery protocol. Different payloads are sent to endpoints subscribed to the topic, based on their delivery protocol.

Message body to send to the endpoint

1	Testing Message From SNS
---	--------------------------

1:25 Text

Message attributes [Info](#)

Message attributes let you provide structured metadata items (such as timestamps, geospatial data, signatures, and identifiers) for the message.

- Confirm that SNS sends the message to the SQS queue.
- Check the queue and verify that the message is present.
- Click on Send and receive messages and check for messages below.

CloudShell Feedback Console Mobile App

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Amazon SNS > Topics > Test-Topic > Publish message

Amazon SQS > Queues > Test-Queue

Test-Queue [Edit](#) [Delete](#) [Purge](#) [Send and receive messages](#) [Start DLQ redrive](#)

Details [Info](#)

Name Test-Queue	Type Standard	ARN arn:aws:sqs:us-east-1:123456789012:Test-Queue
Encryption Amazon SQS key (SSE-SQS)	URL https://sqs.us-east-1.amazonaws.com/123456789012/Test-Queue	Dead-letter queue -

[More](#)

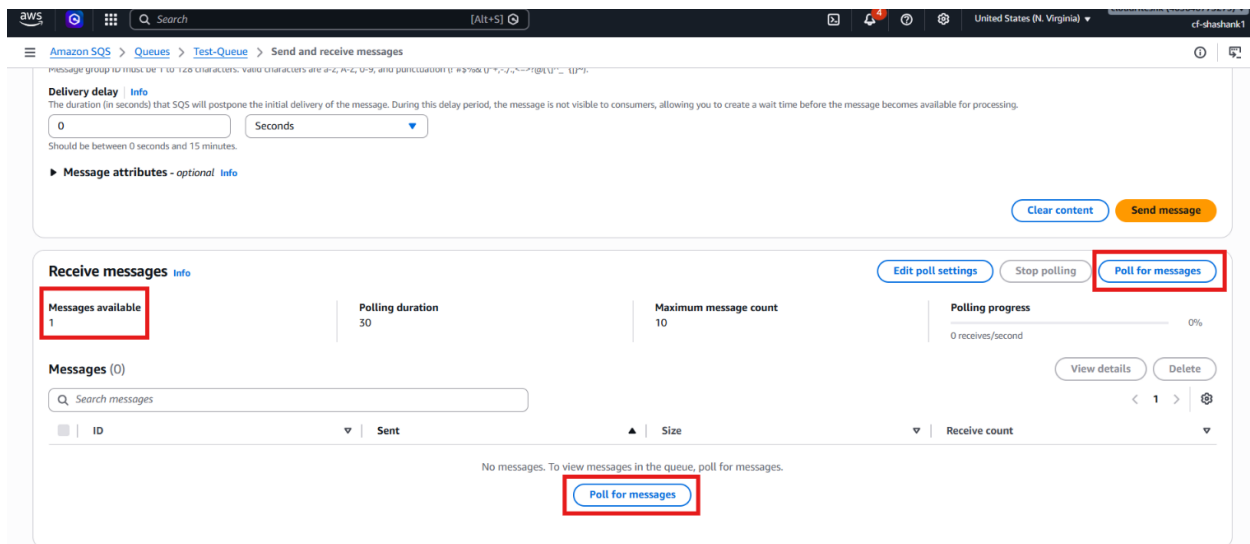
[Queue policies](#) [Monitoring](#) [SNS subscriptions](#) [Lambda triggers](#) [EventBridge Pipes](#) [Dead-letter queue](#) [Tagging](#) [Encryption](#) [Dead-letter queue](#)

Access policy [Info](#) [Edit](#)

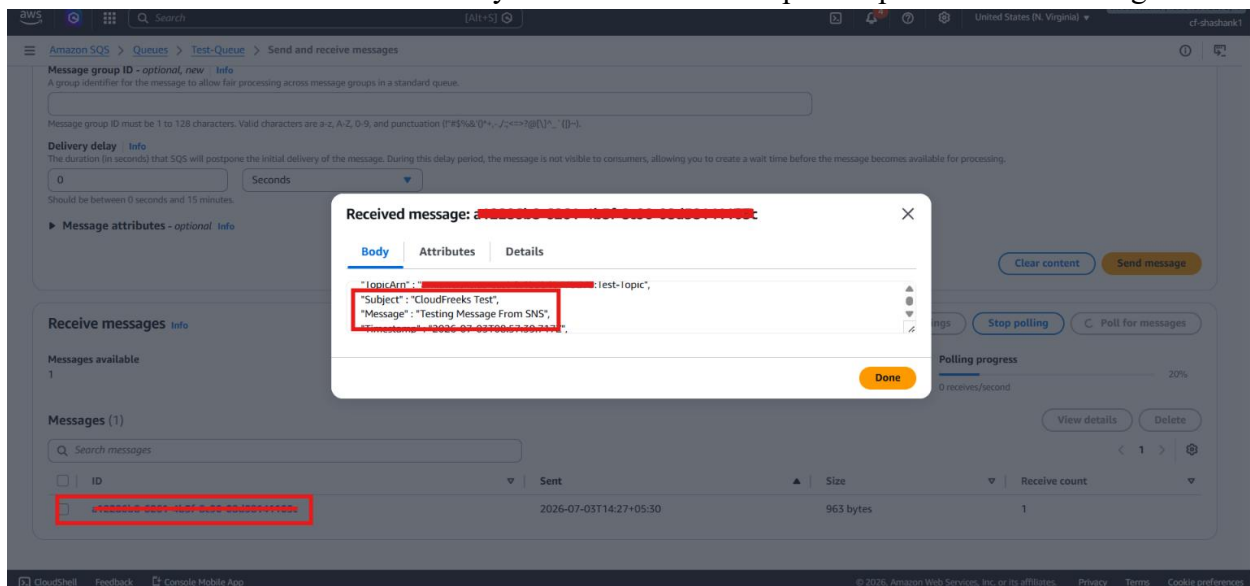
Define who can access your queue.

```
{
  "Version": "2012-10-17",
  "Id": "_default_policy_ID",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": "*",
      "Action": "sqs:*",
      "Resource": "*"
    }
  ]
}
```

- If you see 1 on Messages available, click on **Poll for messages** to see the messages.



- Now click on the most recent one you have received to open it up and see the message.



- Confirm that the queue policy now allows SNS delivery.

Verification

- A standard SNS topic was created successfully.
- A standard SQS queue was created successfully.
- The queue was subscribed to the SNS topic.
- The SQS access policy was updated to allow the SNS topic.
- SNS was able to send messages to SQS after the policy change.
- The SNS → SQS integration worked as expected.